Skills Summary

Reading and Sketching Slope Fields

Use this reference while completing your worksheet or when studying.

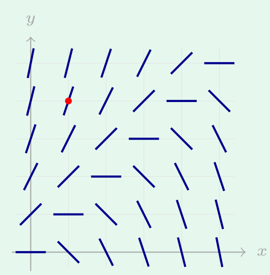
What a slope field is. For $\frac{dy}{dx} = F(x, y)$, the field draws a short segment at each lattice point whose slope is the number F gives there. No solving happens — only substitution. A solution curve is any curve that stays tangent to the segments it passes.

- F has no y in it $\Rightarrow$ each vertical column is one repeated segment.
- F has no x in it $\Rightarrow$ each horizontal row is one repeated segment, and any root of F gives a constant solution.
- Solving $F = 0$ gives the *zero-slope isocline*, the curve on which every segment is flat — it is where solution curves turn.

1. Slope at a lattice point

Substitute the point's coordinates into the right-hand side and stop. The number you get is the slope of the little segment drawn there — not a y-value, not a point. Draw the segment short, and steeper for bigger magnitudes.

Example: For $\frac{dy}{dx} = y - x$, what does the field draw at the point where $x = 1$ and $y = 4$?
Step 1. The right-hand side is the slope itself, so this is substitution, not solving — put the coordinates in and read off the number.
Step 2. $y - x$ becomes $4 - 1$ at that point.



Slope of the segment there: **3** — a steeply rising tick, matching the marked point in the picture.

Try it: For the same equation $\frac{dy}{dx} = y - x$, what slope does the field draw where $x = 3$ and $y = -1$?

check: **−4**

2. Reading structure off a printed field

Two questions unlock almost any printed field. *Does the picture change as you move straight up?* If not, there is no y on the right-hand side. *Where are the flat segments?* Those points solve $F = 0$, and that curve is the zero-slope isocline. Solve it for y so you can draw it.

Example: For $\frac{dy}{dx} = 2x - y$, find the line along which every segment is horizontal.
Step 1. Horizontal means slope zero, so set the whole right-hand side to zero — this is the one place in slope fields where you really do solve.
Step 2. $2x - y = 0$, and solving for y isolates it in one move. Zero-slope isocline: $y = \boldsymbol{2x}$. Solution curves crossing that line are turning there, and the field's flat ticks trace it out.

Try it: For $\frac{dy}{dx} = x - 2y$, find the line of horizontal segments.

check: $y = \boldsymbol{x/2}$

3. The solution curve through a given point

Start at the given point and move so the curve stays tangent to the segments around it; never cross one at an angle, and never let two solution curves touch. When F depends on x alone you can also check yourself with algebra: antidifferentiate, then use the given point to pin down the constant.

Example: $\frac{dy}{dx} = 4x$ with the curve through the point where $x = 0$ and $y = 2$. Find the height of that curve at $x = 1$.
Step 1. The right-hand side has no y, so the equation can be antidifferentiated directly — the sketch and the algebra should agree.
Step 2. An antiderivative of $4x$ is $2x^2$, so $y = 2x^2 + C$; the given point forces $C = 2$, and then $y = 2x^2 + 2$ is evaluated at $x = 1$. Height there: **4**, so the curve leaves the marked point and climbs through that height one unit to the right.

Try it: $\frac{dy}{dx} = 6x$ through the point where $x = 0$ and $y = -1$. How high is the curve at $x = 2$?

check: $y = 3x^2 - 1$, so the height is **11**

(!) Watch out: the number you compute is a *slope*, not a height — at a point where the field says -3, the segment falls steeply; it does not sit at -3. And a constant solution is a genuine solution: no other solution curve may cross it, which is why curves either approach an equilibrium line forever or run away from it, but never touch it.